MIL-B-50015B
17 May 1984
SUPERSEDING
MIL-B-50015A
30 June 1960

MILITARY SPECIFICATION

BULB, ASPIRATOR

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of natural rubber bulbs used to obtain air samples.

1.2 Classification. Aspirator bulbs shall be of the following types as specified (see 6.2):

Type I - 1/8-inch inlet hole
Type II - 3/16-inch inlet hole

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-601 - Rubber, Sampling and Testing

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

FSC 6640

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-177 - Rubber Products, Terms for Visible Defects of
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. The aspirator bulb shall be made of a compound containing pure natural rubber.

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3.1.2 Valve. The valve shall consist of a nickel-plated brass seat with a stainless steel ball.

3.2 Dimensions. The dimensions of the finished bulb shall conform to figure 1.

3.3 Performance.

3.3.1 Air volume. The aspirator bulb, when deflated, shall draw not less than 20 milliliters (mL) of air when tested in accordance with 4.2.4.1.

3.4 Workmanship. The bulb surfaces shall be smooth and free from tackiness, porous areas, cracks, checks, flash, abrasions and blisters. The rubber shall be compact and free from bubbles and foreign matter. Ridges of moldparting lines, when closely trimmed, shall not be considered as flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the aspirator bulbs of one type, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. A lot of bulbs shall include not more than one batch of rubber (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Fifty bulbs shall be taken at random for test.

4.2.3 Inspection procedure.

4.2.3.1 For examination. The sample bulbs taken in 4.2.2.2 and the packaging thereof shall be inspected in accordance with MIL-STD-105 and the following classification of defects. MIL-STD-407 shall apply as a visual inspection guide for rubber-molded items. Defects are as defined in MIL-STD-177.

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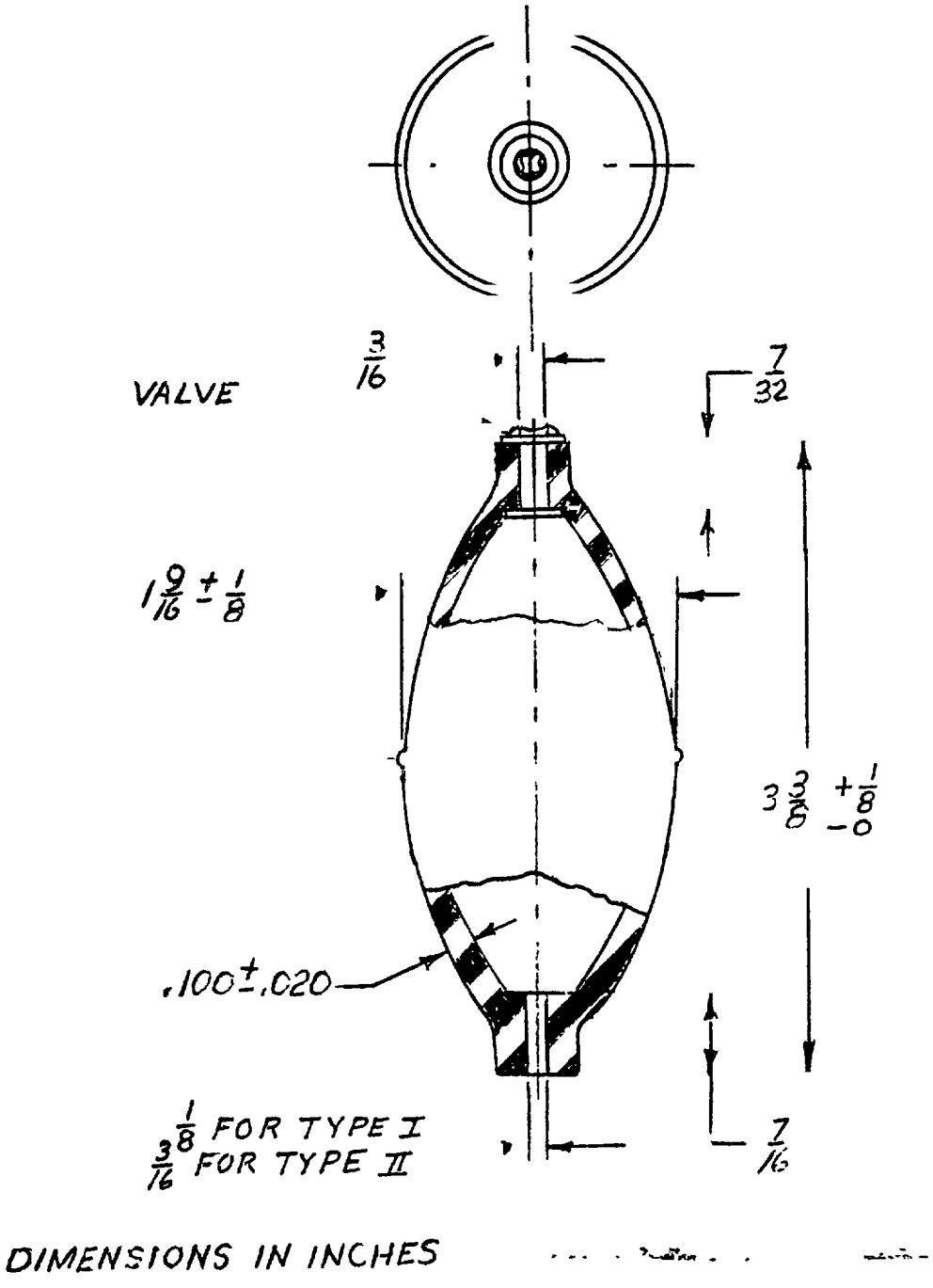


FIGURE 1. Bulb, Aspirator.

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(a) Bulbs, aspirator (see figure 1).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Crack extending through body
102	Blister extending through body
103	Sponginess or porosity
104	Abrasion
105	Valve length incorrect
106	Inlet-hole diameter incorrect
107	Lanyard length incorrect (when required)
108	Contamination (grease or oil)
Minor:	AQL 2.5 percent defective
201	Diameter incorrect
202	Length incorrect
203	Excess flash
204	Flow marks

(b) Packaging (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Contents per container not as specified
102	Container damaged or leaking
103	Unitization not as specified
104	Marking incorrect, missing or illegible

4.2.3.2 For test. Each sample bulb taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests.

4.2.4.1 Air volume. Air volume shall be determined on 50 bulbs before aging and on 25 bulbs after aging as follows: Connect the bottom end of an inverted 50 mL buret (with stopcock open) to a leveling bulb by means of rubber tubing. Place the mouth of the leveling bulb approximately level with the zero graduation of the buret and fill the leveling bulb with water. Raise and lower the leveling bulb several times to expel any trapped air from the rubber tubing,

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then adjust the water level in the buret to the zero graduation. Connect the nozzle of the bulb by means of snug-fitting glass tubing (2 inches or less) and a short piece (12 inches or less) of suitable rubber tubing to the tip of the buret. With the air sampling bulb in a horizontal position and the water level at zero, deflate the bulb completely, then release. Immediately after deflation of the rubber bulb, and before allowing to inflate, raise the leveling bulb so that the water level in the buret is at the 15 mL level. Let the bulb completely inflate, pulling the column of water up into the buret. Determine the volume of air displaced by equalizing the water level in the leveling bulb and buret and read the buret.

4.2.4.2 Aging (oven process). The bulbs selected for oven-process aging shall be aged in an air oven at a temperature of 158° ± 2°F for 168 hours, as specified in FED-STD-601.

5. PACKAGING

5.1 Unit packing, industrial. A quantity of aspirator bulbs, as specified (see 6.2), shall be unit packed industrially in accordance with ASTM D 3951.

5.2 Packing, industrial. Aspirator bulbs, unit packed as specified in 5.1 shall be packed industrially in accordance with ASTM D 3951.

5.3 Unitization. Packs of aspirator bulbs shall be unitized in a manner to assure safe delivery at destination, acceptance by common carrier, and conformance with Uniform Freight Classification Rules and National Motor Freight Classification Rules.

5.4 Marking. Containers shall be marked in accordance with ASTM D 3951 and shall show date of manufacture and lot number of the aspirator bulbs.

6. NOTES

6.1 Intended use. Rubber bulbs covered by this specification are intended for use in detector kits M18A2 and M72A2.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Type of aspirator bulbs required (see 1.2), and
(c) Quantity of aspirator bulbs to be unit packed industrially (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement (ABC-ARMY-STD). When amendment, revision or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.5 Supersession data. Type I bulbs covered by this specification supersede and include the requirements of MIL-B-50015A, dated 30 June 1960. Type II bulbs are a new type.

Custodians:

Army - EA
Air Force - 99

Review activity:

Air Force - 82

Preparing activity:

Army - EA

Project No. 6640-1226

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-B-50015B	BULB, ASPIRATOR

3a NAME OF SUBMITTING ORGANIZATION

b ADDRESS *(Street, City, State, ZIP Code)*

4 TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* -- Optional

b WORK TELEPHONE NUMBER *(Include Area Code)* -- Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* -- Optional

8 DATE OF SUBMISSION *(YYMMDD)*

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